

$\therefore v$  is False. From ③ and defn. of biconditional,  $v$  is False.

$\therefore$  Diana did not access. From ⑤ and defn. of biconditional,  $\neg s$  is False. By defn. of negation,  $s$  is True.

$\therefore p$  must be True. From ① and defn. of bi-conditional,  $t$  is True.

$\therefore$  Carlos did the access

$\therefore q$  must be True. From ② and defn. of biconditional,  $\neg u$  is True.

From defn. of negation,  $u$  is False.  $\therefore$  John did not access

Ans: Carlos did the access

4) Suppose there are signs on the doors to two rooms. The sign on the first door reads "In this room, there is a lady, and in the other one there is a tiger"; and the sign on the second door reads "In one of these rooms, there is a lady, and in one of them there is a tiger." Suppose that you know that one of these signs is true and the other is false. Behind which door is the lady?

Soln:  $p$ : First room has the lady  $q$ : Second room has the lady  
 $r$ : First room has the tiger  $s$ : Second room has the tiger

First room:  $p \wedge s$       Second room:  $(p \wedge s) \vee (q \wedge r)$

~~①~~ ①  $p \Leftrightarrow \neg q$  is True      ②  $r \Leftrightarrow \neg s$  is True

Given, one room speaks the truth and the other lies.  
Let, the first room's sign be true.  $\therefore (p \wedge s)$  is True

By defn. of OR,  $(p \wedge s) \vee (q \wedge r)$  is True, which is the Second room's sign.  
But this is a contradiction to the given constraint.  $\therefore$  First room must be False, i.e.  $(p \wedge s)$  is False. By given constraint, Second room speaks the Truth.

$\therefore (p \wedge s) \vee (q \wedge r)$  is True. By defn. of OR,  $(q \wedge r)$  is True. By defn. of AND,  
 $q$  is True,  $r$  is True.  $\therefore$  First room has the tiger  
Second room has the lady